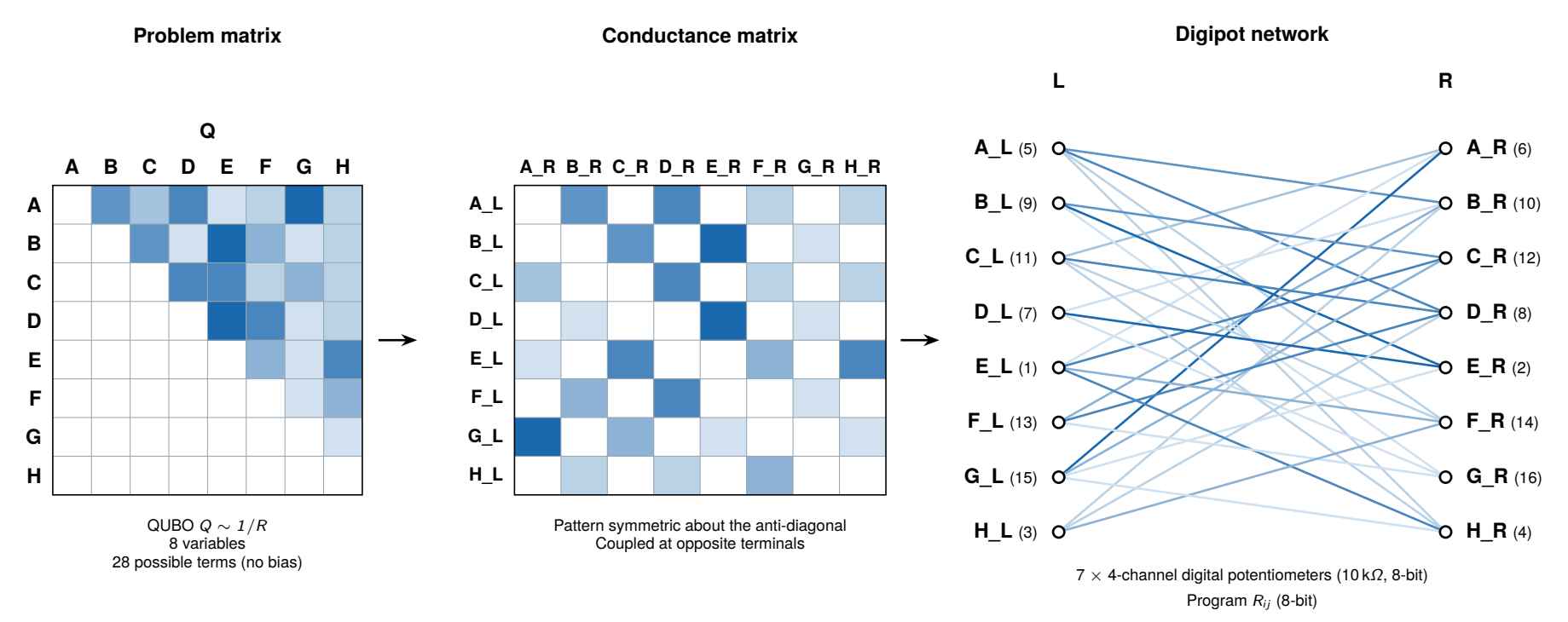
Problem matrix
Q
A B C D E F G H
A
B
C
D
E
F
G
H
QUBO $Q \sim 1/R$
8 variables
28 possible terms (no bias)
Conductance matrix
A_R B_R C_R D_R E_R F_R G_R H_R
A_L
B_L
C_L
D_L
E_L
F_L
G_L
H_L
Pattern symmetric about the anti-diagonal
Coupled at opposite terminals
Digipot network
L
R
A_L (5)
B_L (9)
C_L (11)
D_L (7)
E_L (1)
F_L (13)
G_L (15)
H_L (3)
A_R (6)
B_R (10)
C_R (12)
D_R (8)
E_R (2)
F_R (14)
G_R (16)
H_R (4)
7 × 4-channel digital potentiometers (10 kΩ, 8-bit)
Program R_{ij} (8-bit)